MANASWI KULAHARA

MSc Geo-Informatics | Remote Sensing & Spatial AI Researcher

tel:+91-8700668758 mailto:manaswikulahara8@gmail.com https://linkedin.com/in/manaswi-kulahara-87b352270/
https://github.com/manaswikulahara https://orcid.org/0009-0008-8480-6218

PROFILE

Geospatial Data Sciencegraduate student with strong experience in Python-based spatial analysis, remote sensing, and GeoAI workflows. Hands-on experience with satellite imagery, GIS libraries, and Jupyter notebooks, with a focus on data validation, model evaluation, and reproducible **geospatial pipelines.** Interested in building and testing reliable **geospatial AI tools and products.**

EDUCATION

Master of Science in Geo-Informatics

TERI School of Advanced Studies

2024 – 2026 | New Delhi, India

- GPA: 3.89/5.0 (77.8%) [Current, after 2 semesters]
- Coursework: Advanced GIS, GNSS, Remote Sensing Applications, Environmental Statistics, Spatial Modeling.
- **Projects:** Hydrological and Hydrodynamic Study of Martian Paleo-lakes: A Case Study of Landon Valley [Minor Project].

Bachelor of Arts (Hons) in Geography

University of Delhi

2021 – 2024 | New Delhi, India

- GPA: 4.00/5.0 (80%)
- Coursework: GIS, Remote Sensing, Statistics, Disaster Management, Bio-Geography, Agriculture and Food Security, Demography and Population Studies, Climatology, Geomorphology, Environmental Science, Statistics.
- **Projects:** Assessment of Disaster Risk in Manali (India): Vulnerability, Impact and Mitigation (2024) [Major Project Part1], *Flood Vulnerability of Yamuna Floodplains: A Case Study of Majnu Ka Tila, Delhi,(India) (2024)* [Major Project Part2], Post-COVID Revival of Tourism: A Case Study of Udaipur,(India) (2023) [Minor Project].

Spanish Language Certificate (A2)

St. Stephen's College, DU

2021 – 2022 | New Delhi, India

- GPA: 4.39/5.0 (87.8%)
- Coursework: Foundational Grammar, Practical Vocabulary, Conversational Skills, Reading & Writing.

12th CBSE

Kendriya Vidyalaya

2021 | New Delhi, India

- GPA: 4.33/5.0 (86.6%)
- Coursework: Geography, Economics, History, English, Physical Education.

10th CBSE

Kendriya Vidyalaya

2019 | New Delhi, India

- GPA: 4.00/5.0 (80%)
- Coursework: Science, Mathematics, Social Sciences, English.

RESEARCH EXPERIENCE

Research Intern

Indian Institute of Remote Sensing (IIRS)-Indian Institute of Remote Sensing (ISRO)

Jun 2025 – Aug 2025 | Dehradun, India

- **Project: Hydrological and Hydrodynamic Study of Martian Paleo-lakes: A Case Study of Landon Valley**

Performed a detailed morphometric analysis of Martian terrain using MOLA–HRSC DEMs, including terrain preprocessing, stream network extraction, and breach modeling to infer paleo-flow conditions. Executed 2D hydrodynamic simulations in HEC-RAS to reconstruct paleo flood events, demonstrating skills in the hydrodynamic modeling techniques required for model validation.

Research Intern

National Institute of Disaster Management (NIDM), Ministry of Home Affairs (Govt.of India)

Oct 2024 – May 2025 | New Delhi, India

- **Project: Spatial Analysis of Fire Incidents and The Impact of the Urban Heat Island Effect in Delhi (2022–2024).**

Led a geospatial risk assessment of over 500 fire incidents, developing spatial models that improved emergency response time analysis by 20% and provided insights into the vulnerability of urban populations.Conducted a geospatial analysis of over 500 fire incidents to understand the spatial patterns of risk and provide insights into the vulnerability of different urban populations. My work on this project culminated in a comprehensive report submitted to the Government of India, outlining the critical need for proactive, community-focused resilience strategies.

Research Intern

University of Virginia (Remote)

May 2024 – Jul 2024 | Charlottesville, USA

- **Project: Can AI See What We Can't? Harnessing Deep Learning and Multi-Temporal Satellite Data for Enhanced Crop Type Mapping and Yield Prediction.**

Developed a robust dataset for AI-driven crop classification by integrating multi-temporal, multi-source EO data, including Sentinel-2 and MODIS. To build a robust dataset for accurate agricultural landscape delineation. Designed and implemented the geospatial framework for the project, applying spatial modeling and map development techniques to enhance AI-driven crop classification and yield forecasting. Conducted spatial analyses to uncover regional crop variations and intercropping complexities, improving the transferability and performance of predictive models.

Research Intern

DS Kothari Centre for Research and Innovation in Science Education, Miranda House, University of Delhi

May 2023 – Jul 2023 | New Delhi, India

- **Project: Micro-shed Delineation and Ecological Assessment: Churachandpur, Manipur.**

Performed a comprehensive watershed management assessment by delineating micro-sheds and performing stream ordering using Cartosat DEM, while analyzing vegetation health through NDVI derived from USGS Landsat data to support sustainable ecological planning.

PUBLICATIONS

Peer-Reviewed Journal Articles

- **IEEE Access (2025) (SCI IF = 3.6):** A CNN-Based Framework for Geometric Alignment of Historical and Satellite Imagery.

Peer-Reviewed Conference Proceedings

- **IGARSS 2025 (Core Rank C):** Can We Predict the Unpredictable? Leveraging DisasterNet-LLM for Multimodal Disaster Classification.

- **IGARSS 2025 (Core Rank C):** How Can Multimodal Remote Sensing Datasets Transform Classification via SpatialNet-ViT?.
- **ICASSP 2025 (Core Rank B):** Can AI See What We Can't? Harnessing Deep Learning and Multi-Temporal Satellite Data for Enhanced Crop Type Mapping and Yield Prediction.
- **ASONAM 2025 (Core Rank B):** Can We Predict Your Next Move Without Breaking Your Privacy?.

CERTIFICATIONS

• **Supervised Machine Learning: Regression and Classification** (Sept 2025)
Issued by: DeepLearning.AI & Stanford Online (via Coursera).

• **Spatial Analysis and Satellite Imagery in a GIS** (May 2025)
Issued by: University of Toronto (Canada)

• **Python for Data Science, AI & Development** (March 2025)
Issued by: IBM Developer Skills Network (via Coursera)

WORKSHOPS ATTENDED

• **Cryosphere & Climate Change Studies**
National Institute of Hydrology, Roorkee, Ministry of Jal Shakti, Govt. of India (23-28 March 2025).

• **Advanced Geocomputational Techniques in Natural Hazards and Disaster Management**
Amity University (Noida, India) and NIDM (Delhi, India) (28-29 February 2025).

• **Capacity Building Workshop on UAV/Drone Application in Environment and Natural Resource Management**
TERI School of Advanced Studies (Delhi, India) (10-12 December 2024).

• **Training Course on Conservation and Management of Inland Water Bodies**
National Institute of Hydrology, Roorkee, Ministry of Jal Shakti, Govt. of India (August 5-9, 2024).

• **Enabling Cities for Disaster Resilience to Achieve Sustainable Development**
Urban Studies and Research Center, Miranda House, University of Delhi(India), in collaboration with National Institute of Disaster Management, National Institute of Urban Affairs, and Delhi Development Management Authority (Govt. of India) (8-12 July 2024).

TECHNICAL SKILLS

Programming- Python (NumPy, Pandas, GeoPandas, Rasterio), C, HTML/CSS, JavaScript, Leaflet, Google Earth Engine,PyTest (basic testing & validation).

GIS Software-QGIS, ArcMap, ArcGIS Pro, ERDAS Imagine, ENVI, SAR (Sentinel-1), Optical (Sentinel-2, Landsat, MODIS).

Geospatial Artificial Intelligence (GeoAI)- Deep Learning (CNNs, Vision Transformers), Graph Neural Networks (Interest), Remote Sensing & Earth Observation.

Cloud & Platforms- Google Earth Engine, GEE API, COG.

Remote Sensing- Sentinel, MODIS, Landsat, LULC Classification, NDVI.

Hydrodynamic Modeling- TUFLOW, HEC-HMS, HEC-RAS 2D and 1D, Flood Risk Mapping.

Databases & Servers- Oracle, PostgreSQL (with PostGIS), GeoServer, SQL.

Tools- Git & GitHub, Linux/Windows, Documentation & technical writing.

RESEARCH INTERESTS

- Development of novel deep learning architectures for multimodal data.
- Representation learning for complex, high-dimensional data (e.g., geospatial, visual).
- Domain adaptation and generalization techniques.
- Computational approaches to understanding forest dynamics under climate change.
- Hydrologic Statistics and Machine Learning for Uncertainty Modeling.
- Foundation Models for Geospatial Artificial Intelligence (Geo-Foundation Models).
- Application of AI/ML for Forecast-Informed Decision Making in Hydrology.
- Representation Learning for Geospatial-Visual Data.
- Integration of Remote Sensing and Numerical Models for Environmental Monitoring.

LANGUAGES

English ●●●●●
Hindi ●●●●●
Spanish ●●●○○

MEMBERSHIPS

IEEE Student Member | IGARSS Student Member

REFEREES

Referee 1 (Postgraduate Minor Project Supervisor):
Name: Dr. Ranjana Ray Chaudhuri
Email: ranjana.chaudhuri@terisas.ac.in
Institution Address: TERI School Of Advanced Studies, Vasant Kunj, New Delhi, India.

Referee 2 (Undergraduate Major Project Supervisor):
Name: Dr. Pravin H. Kumar
Email: praveendse@gmail.com
Institution Address: Shyama Prasad Mukherji College for Women, West Punjabi Bagh, New Delhi, India.

Referee 3 (Undergraduate Minor Project Supervisor):
Name: Dr. Rachna Dua
Email: rachnadua123@gmail.com
Institution Address: Shyama Prasad Mukherji College for Women, West Punjabi Bagh, New Delhi, India.

Referee 4 (IIRS-ISRO, Research Intern Supervisor):
Name: Dr. Praveen Kumar Thakur
Email: praveen@iirs.gov.in
Institution Address: IIRS-ISRO, Dehradun, Uttarakhand, India.

Referee 5 (NIDM-Research Intern Supervisor):
Name: Dr. Garima Aggarwal
Email: garima.nidm@govcontractor.in
Institution Address: NIDM, Rohini, New Delhi, India.

Referee 6 (University of Virginia-USA, Research Intern Supervisor):
Name: Dr. Jiechao Gao
Email: jg5ycn@virginia.edu
Institution Address: University of Virginia, Charlottesville, VA, USA.